

MID TERM EXAMINATION
B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
 II Semester, March 2026

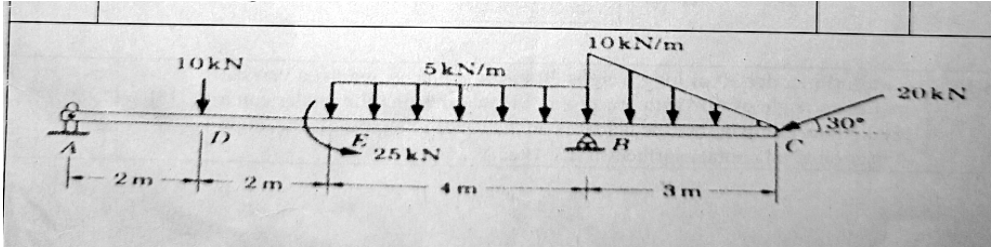
Paper Code: ES 114

Subject: Engineering Mechanics

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question(s)	Max. Marks	CO(s)
1(a)	State Varignon's Theorem and write its mathematical expression.	[2]	1
1(b)	Define the Perpendicular Axis Theorem. For which type of body is it applicable?	[2]	1
1(c)	What is meant by equilibrium? State the conditions of equilibrium for a concurrent coplanar force system.	[2]	1
1(d)	List at least four important assumptions considered while analysing a perfect truss.	[2]	2
1(e)	Explain the terms angle of friction and angle of repose.	[2]	2
Question(s) (Attempt any two)			
2	A beam is hinged at A and roller supported at B. It is acted upon by loads as shown in Figure. Find the reactions at A and B. 	[10]	1

Q. No.	Question(s)	Max. Marks	CO(s)
3	A truss is loaded and supported as shown in figure below. Determine the axial forces in the members CE, CG and FG.	[10]	2
4(a)	Determine the centroid ($\bar{y}$) of the I section for the given figure:	[5]	1
4(b)	A uniform ladder 30 m long weighs 30 kg. It is placed against a vertical wall at an angle of 60° with the ground. How far along the ladder can a 150 kg man climb before the ladder is on the verge of slipping? The angle of friction at all contact surfaces is 15° (i.e. $\phi = 15^\circ$).	[5]	2

**** End of Question Paper ****